

EXERCISES AND EXACT SOLUTIONS

Climate Signals Before Disease Surges

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C28
Notebook	V7_B_N08
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add temperature.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Prevent temporal leakage.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Explain causality limits.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Add dated exposures and nonlinear/lagged terms with domain justification.

Solution 2

Fit baselines and choose parameters only on prior data, then evaluate forward.

Solution 3

Shared seasonality, reporting, interventions, ecology, and confounding can create association; causal claims require stronger design and evidence.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated